WHEN YOU NULL, YOU KNOW:
NATURALNESS JUDGMENTS IN MEXICAN SPANISH CATAPHORA

BY

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ABSTRACT

This study investigates pronoun resolution in Mexican Spanish cataphora, where pronouns precede their possible antecedents. Previous work on anaphora has focused on the Position of Antecedent Strategy, which suggests null pronouns resolve dependencies in favor of subject antecedents, while overt pronouns favor object antecedents. In this study, we use a naturalness judgment task to test whether Mexican Spanish speakers show a preference for resolving null cataphors with subject antecedents. We further investigate whether overt pronouns preferentially resolve to object antecedents, as predicted by a neat division of labor. Broadly, we contrasted an impatient-parser account, where comprehenders aim to resolve dependencies as quickly as possible, with a division-of-labor account. Results suggest that while null cataphors show a clearer sensitivity to subject antecedents, overt pronouns do not show a clear object-antecedent preference. These findings suggest partial support for predictions based on the Position of Antecedent Strategy, and we use this study as a preliminary base for future self-paced reading work on real-time processing.

Keywords: cataphora, naturalness judgment task, pronoun resolution, Position of Antecedent Strategy, Mexican Spanish, psycholinguistics

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Table of Contents

1. Introduction

The interpretation of pronominal elements by comprehenders is crucial for everyday communication. Psycholinguists have long sought to understand the mechanism by which language users reduce ambiguity and maintain discourse coherence. For comprehenders, pronouns pose a unique challenge in establishing interpretable meaning. These syntactic elements are often regarded as 'underspecified,' requiring listeners and readers to rely on syntactic and semantic context, salience, and the prominence of potential referents, among other factors (Bel et al., 2026; Kaiser, 2014; Keating et al., 2011). Pronouns are often preceded by multiple potential antecedents, presenting an ambiguity problem. Despite this, native comprehenders seemingly resolve pronouns efficiently and effectively.

This question of resolution is especially interesting for *pro*-drop languages, which allow pronominal subjects to be omitted. In Spanish, constructions can be formed in which the subject need not be overt. That is to say, the subject does not always need to surface with a lexical noun phrase or pronoun, as illustrated in (1) (Keating et al., 2016).

(1)	a.	Marco/Él	llegó	tarde	a	clase.
		Marco/he	arrive.PST.3SG	late	to	class
		'Marco/he arrived late to class.'				
	b.	∅	llegó	tarde	a	clase.
		PRO.3SG	arrive.PST.3SG	late	to	class
		'He arrived late to class.'				

Prior work on *pro*-drop languages suggests an important contrast between null subjects and their overt counterparts. Namely, the choice of pronoun is often governed by discourse-pragmatic factors that establish interpretive differences. For example, the choice of a null subject often indicates topic continuity, while overt subjects signal a change in a referent. If *pro*-drop speakers use different pronoun forms to indicate a change or continuation of a discourse topic, and comprehenders are sensitive to this choice, then language users may use that same sensitivity to guide pronoun resolution. That is to say, comprehenders may rely on discourse-pragmatic cues, associated with pronouns, to resolve intrasentential ambiguity. Crosslinguistically, it has been suggested that in ambiguous contexts, language users do rely on intuitive biases to establish antecedent-pronominal relationships. Notably, Carminati (2002) examined anaphoric resolution in Italian speakers and proposed the Position of Antecedent Strategy (PAS). Here, anaphora refers to constructions in which the pronoun is preceded by its antecedent. An example is illustrated in (2).

(2) Mariana vio a Daniela bailar cuando ella estaba sola.
Mariana see.PST.3SG DOM Daniela dance when she be.PST.IPFV.3SG alone.F.SG
'Mariana saw Daniela dance when she was alone.'

Her proposal ultimately postulates a division of labor aiming to explain how Italian speakers and comprehenders resolve pronominal elements with possible referents. The PAS suggests that resolution between null and overt pronouns is based on syntactic hierarchy, and in particular, on its relation to the inflectional phrase [Spec, IP]. Thus, taking Carminati's account, the null pronoun would prefer antecedents occupying the specifier of the highest inflectional phrase [Spec, IP]. This contrasts with the preference of the overt pronoun for antecedents sitting lower in the phrase structure. As illustrated in (3a) and (3b), extending the PAS to Spanish speakers would result in an expectation that ambiguity would be resolved systematically by pronoun form: the null pronoun favors the matrix subject position/Spec-IP antecedent, while the overt pronoun favors the matrix object position/non-Spec-IP antecedent (Keating et al., 2016). Hence, for the anaphoric construction in (3a), the felicitous choice for native speakers is in favor of *Elena*, the subject antecedent. By contrast, in (3b), speakers would be biased towards *Isabella*, who occupies the object position.

(3) a. Elena$_i$ visitó a Isabella$_j$ cuando $\varnothing_i$ estaba en la ciudad.
Elena visit.PST.3SG DOM Isabella when pro.3SG be.PST.IPFV.3SG in DEF.F.SG city
'Elena visited Isabella when she was in the city.'

b. Elena$_i$ visitó a Isabella$_j$ cuando ella$_j$ estaba en la ciudad.
Elena visit.PST.3SG DOM Isabella when she be.PST.IPFV.3SG in DEF.F.SG city
'Elena visited Isabella when she was in the city.'

The vast majority of the literature on pronoun interpretation has focused on pronoun resolution in anaphora (4a), where pronouns are introduced after their antecedents. In contrast, there has been very little work examining cataphora or backward anaphora (4b), where pronouns are presented before their antecedents (Kaiser, 2014).

(4) a. Cuando Ricardo canta, él despierta a todos.
when Ricardo sing.PRS.3SG he wake.PRS.3SG DOM everyone
'When Ricardo sings, he wakes everyone up.'

b. Cuando él canta, Ricardo despierta a todos.
when he sing.PRS.3SG Ricardo wake.PRS.3SG DOM everyone
'When he sings, Ricardo wakes everyone up.' (Kaiser, 2014)

Previous studies have investigated the distinction between anaphora and cataphora, revealing the latter's complexity crosslinguistically (Kaiser, 2014). Cataphora, unlike its counterpart, requires that comprehenders anticipate resolving a pronoun before it encounters its antecedent. Notably, whereas anaphoric pronouns develop discourse relationships with previous antecedents, cataphora has been argued to involve "active-search" mechanisms that guide its resolution (Kaiser, 2014; Pavlič & Stepanov, 2023). Thus, extending the work of Carminati's PAS is essential to capturing potential, distinctive, and predictive mechanisms that comprehenders create when encountering a cataphoric pronoun. In this paper, we report naturalness judgments on Spanish cataphora to motivate further investigation of an upcoming self-paced reading study. We investigated acceptability across pronoun type and subject match as an offline baseline to test whether a division of labor (in pronouns) contributes to the human language processing mechanism for an SPR study.

1.1 The division of labor and Spanish pronoun resolution

In Carminati (2002), the Position of Antecedent Strategy was originally formulated for anaphora resolution in Italian. Subsequent work with the PAS views the account as supporting a division-of-labor account. That is, the PAS is often used to describe preferences associated with null and overt pronouns. Under this account, null pronouns are proposed to prefer structurally prominent antecedents, while overt pronouns prefer antecedents lower in the phrase structure. Carminati's

(2002) data strongly suggested a complementary and robust effect of intrasentential anaphora. Follow-up work in other *pro*-drop languages has since attempted to extend the original findings.

In 2011, Keating et al. investigated whether different populations of Spanish speakers utilized distinct assignment strategies in ambiguous anaphora. Across groups of native monolinguals, adult early Spanish-English bilinguals (heritage speakers), and late English-Spanish bilinguals (adult second language [L2] learners of Spanish), Keating et al. (2011) used a forced-choice comprehension task to test how Spanish input affects strategies for anaphora resolution. By presenting globally ambiguous sentences, defined as lacking in pragmatic, grammatical, or contextual biases for either antecedent, participants viewed complex sentences consisting of a main clause followed by a subordinate one. In each item, both potential antecedents were presented by the main clause, while the subordinate clause introduced the pronoun. Crucially, of the 40 critical items, 20 were made up of null pronouns while the other half consisted of overt pronouns. Example stimuli taken from Keating et al. (2011) are represented in (5) and (6).

(5) Daniel ya no ve a Miguel desde que ∅ se casó.
Daniel already no see.PRS.3SG DOM Miguel since PRO.3SG REFL marry
'Daniel no longer sees Miguel ever since he got married.'

(6) Alicia se encontró con Elena mientras ella corría en el parque esta mañana.
Alicia REFL find.PST.3SG with Elena while she run.PST.IPFV in the park this morning
'Alicia ran into Elena while she was running in the park this morning.'

(Keating et al., 2011)

Following each item, participants were presented with a comprehension question eliciting resolution preferences for pronouns in the subordinate clause. A sample question for the item in (5) is depicted in (7).

(7) ¿Quién se casó?
who REFL marry.PST.3SG
'Who got married?'

A. Daniel B. Miguel (Keating et al., 2011)

Their findings are relevant to the question at hand, as their monolingual speakers showed a strong null pronoun-subject bias. Null-pronoun results were consistent with Carminati's (2002) study and the PAS, with more participants selecting the antecedents at the subject position in relation to null pronouns (74%) in comparison to their overt counterparts (37%). However, monolingual speakers in Keating et al. did not provide evidence of a neat division of labor. An aggregate analysis revealed that monolingual speakers interpreted the overt pronoun in favor of object antecedents less than half of the time (46.21%). At an individual level, analysis shows that within the overt condition, only 21% of participants showed an object bias, in contrast to a subject bias in the same condition (37%). Thus, these findings suggest partial support for the PAS in offline measurements, where participants show a clear subject bias with null pronouns, as opposed to weakened subject-bias effects in overt conditions. Additionally, overt pronouns show no clear indication of a bias for objects (Keating et al., 2011).

By contrast, online self-paced reading data offer a different perspective. Results from native Mexican Spanish speakers align more strongly with the original findings of Carminati (2002). Using a self-paced reading task, Keating et al. (2016) examined the resolution of intrasentential anaphora among monolingual and heritage bilingual speakers. Using complex sentences composed of subordinate clauses followed by a main clause, Keating et al. adopted a 2×2 factorial design crossing pronoun type and whether the sentence semantically favors the subject/Spec-IP or object/non-Spec-IP antecedent. Each item introduced two NPs of the same gender in the subordinate clause and utilized the main clause to introduce the pronoun.

Pronouns introduced in the main clause potentially could resolve to either antecedent, but the sentence context provided by the main clause semantically favors one (8).

(8) a. Después de que el sospechoso habló con el policía,
after that DEF.M.SG suspect speak.PST.3SG with DEF.M.SG policeman
∅ admitió su culpabilidad.
PRO.3SG admit.PST.3SG POSS.3SG guilt
'After the suspect spoke with the policeman, he admitted his guilt.'

b. Después de que el policía habló con el sospechoso,
after that DEF.M.SG policeman speak.PST.3SG with DEF.M.SG suspect
∅ admitió su culpabilidad.
PRO.3SG admit.PST.3SG POSS.3SG guilt
'After the policeman spoke with the suspect, he admitted his guilt.'

c. Después de que el sospechoso habló con el policía, él
after that DEF.M.SG suspect speak.PST.3SG with DEF.M.SG policeman he
admitió su culpabilidad.
admit.PST.3SG POSS.3SG guilt
'After the suspect spoke with the policeman, he admitted his guilt.'

d. Después de que el policía habló con el sospechoso, él
after that DEF.M.SG policeman speak.PST.3SG with DEF.M.SG suspect he
admitió su culpabilidad.
admit.PST.3SG POSS.3SG guilt
'After the policeman spoke with the suspect, he admitted his guilt.'

(Keating et al., 2016)

As an additional offline measure and to ensure active engagement, participants were asked to complete comprehension questions in one of two varieties: half were "who"-type as seen in (9), while the other half were "where"-type. "Who"-type constructions aimed to elicit final interpretation of the pronoun in the main clause, in contrast to "where"-type constructions, which ascertained where events likely took place.

(9) ¿Quién admitió su culpabilidad?
who admit.PST.3SG POSS.3SG guilt
'Who admitted their guilt?'

A. El policía B. El sospechoso

'The policeman.' 'The suspect.' (Keating et al., 2016)

Reading-time results from Keating et al. (2016) revealed PAS-like resolution for ambiguous intrasentential constructions. That is to say, participants read clauses containing null subjects significantly faster when resolved in favor of subject antecedents. Conversely, participants read clauses with overt subjects significantly faster when those clauses were resolved in favor of object antecedents.

Overall, the online self-paced reading task suggested that native Mexican Spanish speakers often link null pronouns to subject antecedents, whereas overt pronouns were linked with object antecedents. Despite some inconsistencies, secondary analysis of residual RTs further corroborated evidence for the PAS. Regardless of the findings in the online task, the offline results of monolinguals reflected a general preference for subject antecedents across both pronoun types (Keating et al., 2016).

In recent work, Catalan further suggests variability with overt-pronoun resolution. Bel et al. (2026) report that even in another *pro*-drop language, null pronouns show a strong preference for resolving ambiguity to subject antecedents. However, among Catalan speakers, resolution strategies differed with overt pronouns. That is to say, they did not fully align with the expectations postulated by the PAS. In particular, overt pronouns did not show clear preferences for object antecedents. If a strong PAS account holds in *pro*-drop languages, then predictions in anaphoric resolution would suggest a clean division of labor between null and overt pronouns. Namely, null pronouns should favor the subject antecedent, whereas overt pronouns should reliably favor object antecedents. Online eye-tracking results suggest that Catalan speakers do show a robust subject bias towards null pronouns. Conversely, speakers show hesitation and delayed resolution in ambiguous overt-pronoun conditions. This observation by Bel et al. ultimately does not align with stronger interpretations of a division of labor, as observations support a prediction demonstrating a null-subject bias but not an overt-object bias.

1.2 Cataphora and active search

Syntactically, anaphora and cataphora both differ in directionality; in addition, comprehenders also resolve their pronoun-antecedent dependencies uniquely. During resolution, anaphora resolves its pronoun with previously presented antecedents. By contrast, cataphora raises an important question about its ability to look forward. Namely, it asks whether comprehenders either utilize "active-search" mechanisms or instead rely on predictive strategies.

Foundational work on cataphora processing sought to answer this same question. Kazanina et al. (2007) examined whether active search mechanisms in cataphoric dependencies are grammatically constrained. Looking at similar long-distance dependencies, such as wh-dependencies, the authors asked whether the same mechanisms could be extended to cataphora. In their first experiment, they conducted an online self-paced reading task using a gender-mismatch paradigm to test the impact of Principle C during active search for pronoun antecedents. Each item had four main conditions and an additional fifth name-control condition, shown in (10). The four main conditions were organized using a 2×2 factorial design on constraint (Principle C vs. no-constraint) and gender congruency (Match vs. Mismatch). Match/Mismatch refers to whether the cataphoric pronoun and the subject of the second clause matched in gender.

(10) a. [Principle C, Match]
Because last semester she$_i$ was taking classes full-time while Kathryn was working two jobs to pay the bills, Erica$_i$ felt guilty.

b. [Principle C, Mismatch]
Because last semester she$_i$ was taking classes full-time while Russell was working two jobs to pay the bills, Erica$_i$ felt guilty.

c. [No constraint, Match]
Because last semester while she$_i$ was taking classes full-time Kathryn$_i$ was working two jobs to pay the bills, Russell never got to see her.

d. [No constraint, Mismatch]
Because last semester while she$_i$ was taking classes full-time Russell was working two jobs to pay the bills, Erica$_i$ promised to work part-time in the future.

e. [No constraint, Name]
Because last semester while Erica$_i$ was taking classes full-time Russell was working two jobs to pay the bills, she$_i$ promised to work part-time in the future.

(Kazanina et al., 2007)

Under the study's design, the parser would be expected to follow Principle C, thus showing no mismatch effect. Importantly, Principle C would block the second subject from being grammatically available. In contrast, in the no-constraint condition, the second subject is a grammatically available potential antecedent for the pronoun. Thus, in (10d), the availability of the second subject would result in slower reading times due to gender mismatch, as the parser would attempt to resolve a cataphoric dependency when possible. Ultimately, Kazanina et al. (2007) found the expected mismatch effect when the second subject was grammatically available. These findings suggest that for cataphora, "active search" is constrained by grammatical constraints such as Principle C.

Pavlič and Stepanov (2023) further extend the question to null cataphora in Slovenian by examining active-search mechanisms in cataphor number-mismatch effects. In this study, they suggested that if a parser links cataphoric pronouns to upcoming antecedents, then a number-mismatch effect should arise when encountering mismatched features. Such a mismatch effect would suggest that the parser previously anticipated or built an expectation. The findings showed online mismatch effects. Thus, the results provide evidence that even in *pro*-drop languages, an "active-search" mechanism is present for cataphoric constructions.

Additionally, Dutch cataphora provides evidence for a more abstract prediction. Giskes and Kush (2023) show that comprehenders use morphosyntactic information from cataphors to generate predictions of an upcoming NP. Yet, as a non-*pro*-drop language, results cannot be extended to null/overt division-of-labor accounts. This raises the question of what guides the active search for cataphoric pronouns. In earlier studies, Cowart and Cairns (1987) suggested that cataphoric pronouns show a strong preference for resolving dependencies with first-antecedents. Subsequent work by van Gompel and Liversedge (2003) built on this finding by using eye-tracking methods, where eye-movement patterns suggested a preference for first nouns over second nouns. Together, these findings suggest a theory in which parsers are "impatient" and try to resolve pronouns as early as they can. Such an account differs from a division-of-labor account. For example, in a construction like (11), a division of labor predicts that pronoun form guides antecedent choice (11a). By contrast, an impatient-parser account would suggest a general preference for first antecedents (11b). The following example focuses on overt pronouns, as they most clearly show a contrast between the two proposals, since null pronouns often point toward subjects/first antecedents.

(11) a. Cuando él$_i$ está feliz, Geraldo llama a Juan$_i$.
when he be.prs.3sg happy.m.sg Geraldo call.prs.3sg dom Juan
'When he is happy, Geraldo calls Juan.'

b. Cuando él$_i$ está feliz, Geraldo$_i$ llama a Juan.
when he be.prs.3sg happy.m.sg Geraldo call.prs.3sg dom Juan
'When he is happy, Geraldo calls Juan.'

In 2014, Fedele & Kaiser extended work on Italian with promising results for the PAS in anaphoric constructions, finding that null pronouns generally prefer subject antecedents, while overt pronouns prefer object antecedents. However, these results were modulated in cataphoric constructions, with both null and overt pronouns strengthening preferences for subjects/first-antecedents and weakening preferences for overt objects. These findings suggest that cataphoric processing may be associated with "active-search" or "impatient" processing strategies. Thus, cataphora may trigger active searches for upcoming antecedents but may also create a general bias favoring subjects and first available antecedents, which competes with accounts based on a division of labor.

1.3 Offline vs. online divergence

Patterns in cataphoric resolution often differed across various studies. In initial offline measures of Mexican Spanish, interpretation preference tasks revealed a statistically significant null-subject bias in monolinguals, but a weakened subject-bias and no object bias in overt conditions (Keating et al., 2011). Later online measures of self-paced reading revealed that Mexican Spanish monolingual speakers appear to have a strong null-subject bias but a weak object bias (Keating et al., 2016).

Differences in offline and online patterns also appear in the only previous study comparing null and overt cataphora. In Korean, cataphora patterns show a similar offline-online split, with offline questionnaires reflecting a subject preference for null pronouns and overt pronouns to objects. However, despite initial findings, these patterns fail to replicate in online measurements with Korean speakers showing a general subject bias with both types of cataphoric pronouns (Kwon & Lee, 2026). The variability in pronominal biases may suggest that across different languages, different methods do not always align with online processing patterns. Thus, findings in the present study should be approached with caution, using the results as a baseline for future online measurements and additional research.

1.4 Present Study

The present study investigates the naturalness judgments of Spanish cataphoric stimuli as a baseline for a future self-paced reading study on predictive processing. The current study seeks to answer two main questions. The first asks whether offline judgments show a basic preference for null cataphora, which favors subject resolutions, and whether those predictions are guided by impatience or a division-of-labor account. The second question asks if these same patterns extend to overt cataphora as predicted by a division-of-labor account.

We investigated these two questions by conducting a naturalness judgment task through Prolific and PCIbex, manipulating (i) pronoun type (null vs. overt) and (ii) subject match (match vs. mismatch).

2. Method

2.1 Participants

Thirty-two native Spanish speakers were recruited online through Prolific ($M_{\text{AGE}} = 28.56$; $SD = 4.81$). They spent approximately 30 minutes completing the experiment for a rate of $15/hour. All participants self-identified as native Mexican Spanish speakers and were born and currently reside in Mexico. Participants were selected between the ages of 18 and 40, with at least a high school degree. Selection criteria also required that participants have previously completed at least 10 Prolific assignments with a 90% approval rating.

2.2 Materials

The materials for the offline task consisted of 36 critical items and 40 filler items. All 36 critical items followed a 2×2 factorial design manipulating Pronoun Type (null vs. overt), and Subject Match (Match vs. Mismatch). The critical items were cataphoric constructions consisting of a main clause followed by a subordinate clause. The main clause introduced the pronominal element, followed by two possible antecedents. Subordinate clauses were headed by complementizers such as *cuando*, *aunque*, *ya que*, and *si*. Half of the clauses contained a null pronoun, and the other half contained an overt pronoun. Within the overt condition, the pronoun in the main clause varied in plurality and gender, consisting of *él*, *ella*, *ellos*, and *ellas*. Examples of the four conditions are provided in (12). Subscripts indicate intended coreference and were not included in the stimulus seen by participants. Match refers to cases where the pronoun's features match the subject antecedent and indicate coreference with that subject. Mismatch refers to cases where the pronoun's features do not match the subject antecedent and instead indicate coreference with the object antecedent.

(12) a. [Null, Match]

Cuando $\varnothing_i$ están contentas, las hermanas$_i$ chismosas llaman muchas
when PRO are.3PL happy.F.PL DEF.F.PL sisters gossipy.F.PL call.3PL many
veces al tío$_j$ soltero.
times OBJ.DEF uncle single.M.SG

'When they are happy, the gossipy sisters call the single uncle many times.'

b. [Null, Mismatch]

Cuando $\varnothing_i$ está contento, las hermanas$_j$ chismosas llaman muchas
when PRO is.3SG happy.M.SG DEF.F.PL sisters gossipy.F.PL call.3PL many
veces al tío$_i$ soltero.
times OBJ.DEF uncle single.M.SG

'When he is happy, the gossipy sisters call the single uncle many times.'

c. [Overt, Match]

Cuando ellas$_i$ están contentas, las hermanas$_i$ chismosas llaman muchas
when they are.3PL happy.F.PL DEF.F.PL sisters gossipy.F.PL call.3PL many
veces al tío$_j$ soltero.
times OBJ.DEF uncle single.M.SG

'When they are happy, the gossipy sisters call the single uncle many times.'

d. [Overt, Mismatch]

Cuando él$_i$ está contento, las hermanas$_j$ chismosas llaman muchas
when he is.3SG happy.M.SG DEF.F.PL sisters gossipy.F.PL call.3PL many
veces al tío$_i$ soltero.
times OBJ.DEF uncle single.M.SG

'When he is happy, the gossipy sisters call the single uncle many times.'

The 40 filler items in this study contained a variety of items. To ensure active participation, 16 were simple constructions, half of which served as comprehension checks by including tense and agreement errors. Below are examples of the two varying degrees of grammatical and ungrammatical constructions (13).

(13) a. [Grammatical]

El talentoso DJ está molesto, por eso ignoró a
DEF.M.SG talented.M.SG DJ be.PRS.3SG upset.M.SG for that ignore.PST.3SG DOM
la gerente.
DEF.F.SG manager

'The talented DJ is upset, that's why he ignored the manager.'

b. [Ungrammatical]

*La sabio científica dirige un laboratorio, entonces ellos
DEF.F.SG wise.M.SG scientist direct.PRS.3SG INDEF.M.SG laboratory so they
está ocupada.
be.PRS.3SG busy.F.SG

Intended: 'The wise scientist runs a lab, so she is busy.'

(13a) was expected to be easy to comprehend and receive a higher rating closer to "7" on a Likert scale from "Muy Raro"/Very Odd to "Muy Natural"/Very Natural. The construction in (13b) would not be expected to be ranked as highly as it contains several agreement errors. Participants would be expected to rate these constructions closer to a "3". We draw this conclusion because all participants completed 4 practice trials before the main trials. After completing practice item 3 (14a), participants received the message shown in (14b).

(14) a. Los ranas brinca si alguien camina cerca de su
DEF.M.PL frog.F.PL jump.PRS.3SG if someone walk.PRS.3SG near of POSS.3SG
habitación.
room.F.SG
'The frogs jump if someone walks near their habitat.'

b. El español es un idioma que codifica género y pluralidad en sus sustantivos.

Por lo tanto, el ejemplo presentado es atípico y probablemente un "3."

'Spanish is a language that encodes gender and plurality in its nouns. Therefore, the example presented is atypical and probably a “3”.’

Additionally, 24 of the 40 filler items were object-relative clause constructions for a planned experiment on interference effects in Spanish object relative clauses. These sentences used a 2×2 factorial manipulation on Relative Clause Type (object vs. subject) and Noun Similarity (similar vs. dissimilar) as depicted in (15).

(15) a. [Object, Similar]

El	niño	que	la	niña	vio	ayer	por	la	tarde
DEF.M.SG	boy	REL	DEF.F.SG	girl	see.PST.3SG	yesterday	by	DEF.F.SG	afternoon

caminaba	por	el	parque.
walk.PST.IPFV.3SG	through	DEF.M.SG	park

‘The boy whom the girl saw yesterday afternoon walked by the park.’

b. [Object, Dissimilar]

El	niño	que	Rebeca	vio	ayer	por	la	tarde
DEF.M.SG	boy	REL	Rebeca	see.PST.3SG	yesterday	by	DEF.F.SG	afternoon

caminaba	por	el	parque.
walk.PST.IPFV.3SG	through	DEF.M.SG	park

‘The boy whom Rebeca saw yesterday afternoon walked by the park.’

c. [Subject, Similar]

El	niño	que	vio	a	la	niña	ayer	por	la	tarde
DEF.M.SG	boy	REL	see.PST.3SG	DOM	DEF.F.SG	girl	yesterday	by	DEF.F.SG	afternoon

caminaba	por	el	parque.
walk.PST.IPFV.3SG	through	DEF.M.SG	park

‘The boy who saw the girl yesterday afternoon walked by the park.’

d. [Subject, Dissimilar]

El	niño	que	vio	a	Rebeca	ayer	por	la	tarde
DEF.M.SG	boy	REL	see.PST.3SG	DOM	Rebeca	yesterday	by	DEF.F.SG	afternoon

caminaba	por	el	parque.
walk.PST.IPFV.3SG	through	DEF.M.SG	park

‘The boy who saw Rebeca yesterday afternoon walked by the park.’

In total, participants read 76 main sentences, including 36 critical trials and 40 filler trials. With the four practice trials, participants completed 80 sentences overall. Each sentence was followed by a

Likert scale eliciting participants' judgments.

2.3 Procedure

Participants completed the naturalness judgment task through Prolific. Stimulus presentation and recording of response choice were conducted with PCIbex. All sentences appeared individually on a single line. Participants used the space bar to advance to a Likert scale (1 - 7) and answered the naturalness of the sentence with 1 coded as "Muy Raro" (Very Odd) and 7 as "Muy Natural" (Very Natural). Experimental items were preceded by instructions written in Spanish and four practice items. All sentences were randomized except the practice sentences before the start of the experiment. Participants were asked to read as naturally as possible and rate using their honest intuition. Participants took around 30 minutes to complete the naturalness judgment task.

2.4 Data Analysis

In analyzing the naturalness ratings of our critical items, we implemented an ordinal regression model using the cumulative link mixed model (CLMM) function from the ordinal package in R. Our main analysis for this study focused on two fixed-effect factors: Pronoun Type and Subject Match. Each factor consisted of two levels: Pronoun Type compared null pronouns and overt pronouns, while Subject Match compared Match and Mismatch conditions. Both factors were sum-coded with values of -0.5 or 0.5, centering the effects around the grand mean. The model also included the interaction between Pronoun Type and Subject Match, which tested whether the effect of Subject Match varied within null and overt pronoun conditions. Additionally, the model included random intercepts for both items and subjects, as well as random slopes for Pronoun Type, Subject Match, and their interaction by subjects and items. The ordinal package was used to obtain relevant p-values from Wald z-tests, and we conducted pairwise post hoc comparisons with the emmeans package to examine factorial effects within each level. Specifically, we looked for the effect of subject match across null and overt pronouns, as well as the effect of pronoun type across match and mismatch conditions. Before our main analysis, we reviewed participants' self-reported primary language and their performance on filler sentences targeting attention and sensitivity to grammaticality. We also inspected all items for atypical behavior relative to the overall pattern.

3. Results

The results shown below in Figure 1 represent the distribution of naturalness ratings across all four conditions.

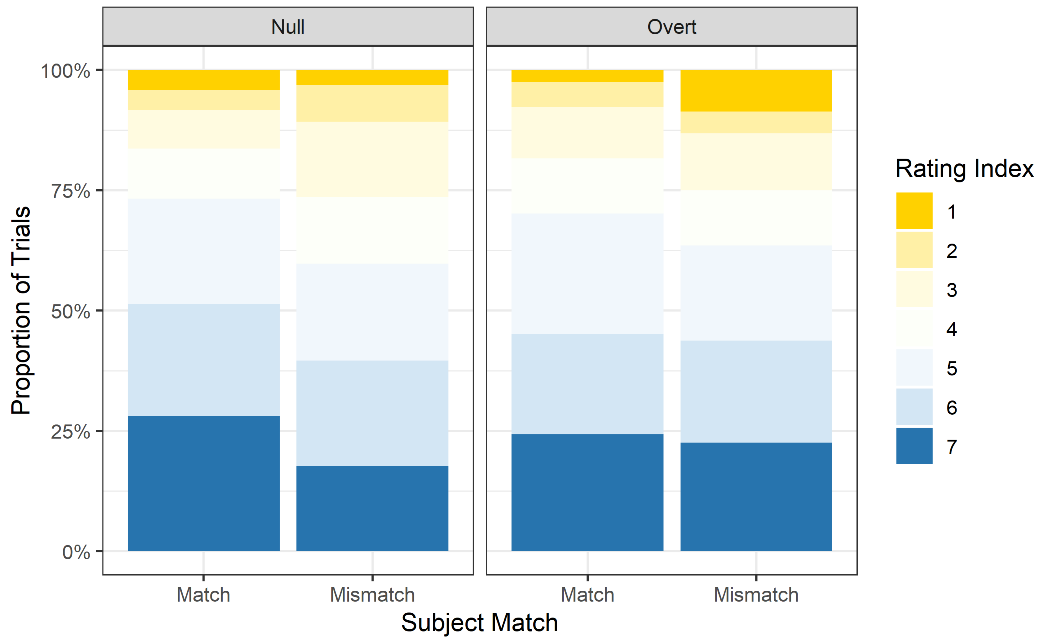

Figure 1: Naturalness Rating Distribution

Overall, the raw distributions show visibly higher naturalness ratings in the Subject Match condition within null pronouns, whereas the Mismatch condition receives fewer ratings of “7”. This contrasts with the overt-pronoun condition, which visually shows a less clear match-mismatch difference. Across null and overt pronouns, the Match conditions show visible similarity in overall naturalness ratings. By contrast, the Mismatch conditions differ, with a slight descriptive advantage for overt mismatches compared to null mismatches.

To further assess these patterns statistically, we performed an ordinal regression analysis using a cumulative link mixed model of the naturalness ratings as reported in Table 1. Our model included fixed effects of Pronoun Type, Subject Match, and their interaction, with random intercepts for participants and items. We also included random slopes by participant and item for Pronoun Type, Subject Match, and their interaction. Analysis of the naturalness ratings revealed a significant main effect of Subject Match, indicating that Match and Mismatch conditions differed in naturalness ratings. In contrast, there was no significant main effect of Pronoun Type, and the Pronoun Type × Subject Match interaction was not significant.

Table 1 Results from Cumulative Link (naturalness ratings). This table reports coefficient estimates, standard errors, z-values, and p-values for the fixed effects.

Table 1: Results from cumulative link mixed model of naturalness ratings.

Predictor	Estimate	SE	z	p
Pronoun Type	-0.023	0.091	-0.257	.767
Subject Match	-0.284	0.107	-2.660	.008
Pronoun Type × Subject Match	0.207	0.168	1.227	.220

Table 2 shows pairwise post hoc comparisons indicating that within null pronouns, Match conditions received higher naturalness ratings than Mismatch conditions (Null: $\beta = 0.388$, SE = 0.125, $z = 3.094$, $p = .002$). However, for overt pronouns, Match and Mismatch did not significantly differ (Overt: $\beta = 0.181$, SE = 0.146, $z = 1.239$, $p = .215$). For Match conditions, Null and Overt did not significantly differ (Match: $\beta = 0.127$, SE = 0.099, $z = 1.283$, $p = .200$). For Mismatch conditions, Null and Overt did not significantly differ (Mismatch: $\beta = -0.080$, SE = 0.145, $z = -0.553$, $p = .580$).

Table 2: Pairwise post hoc comparisons.

Condition	Estimate	SE	z	p
Null Match – Mismatch	0.388	0.125	3.094	.002
Overt Match – Mismatch	0.181	0.146	1.239	.215
Match Null – Overt	0.127	0.099	1.283	.200
Mismatch Null – Overt	-0.080	0.145	-0.553	.580

Figure 2 shows the model-estimated patterns across Pronoun Type and Subject Match conditions. Consistent with the post hoc analysis found in Table 2, the model estimates that within the null-pronoun condition, Null Match conditions receive higher estimated naturalness ratings than Null Mismatch conditions. This reflects our pairwise post hoc comparisons, which reveal a significant Match-Mismatch difference for null pronouns. In contrast, while the overt-pronoun condition shows a small descriptive Match-Mismatch difference, our pairwise comparisons did not detect a significant Match-Mismatch difference within overt pronouns. Moreover, the interaction between Pronoun Type and Subject Match was not statistically significant, so the interpretation of the observed descriptive difference should be taken with caution.

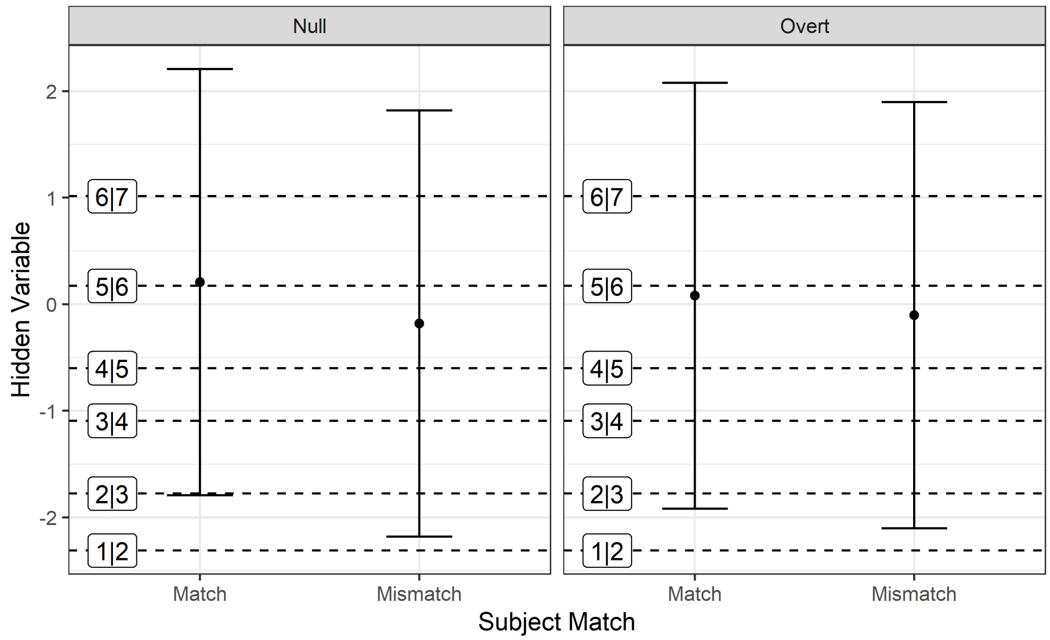

Figure 2: Model-estimated naturalness patterns across Pronoun Type and Subject Match.

In summary, offline judgments of cataphoric null and overt pronouns suggest sensitivity to subject match for null pronouns. However, statistical models reveal no significant Match-Mismatch difference for overt pronouns. Additionally, our models showed no significant interaction between Pronoun Type and Subject Match. Despite the lack of a significant interaction, the significant main effect of Subject Match suggests that participants show some sensitivity to pronoun-antecedent compatibility. However, because these judgments come from an offline task, they may reflect final interpretations rather than real-time processing. For this reason, these patterns should be further examined with an online measure.

4. Discussion

The goal of this study was to investigate whether null cataphors show subject-antecedent preferences in offline judgments. Additionally, we asked whether those patterns extended to overt cataphors and object antecedents as proposed by a strong division of labor. More broadly, we aimed to assess whether cataphoric resolution preferences are guided by impatience or a division-of-labor account. We tested these predictions using a naturalness judgment task with cataphoric null and overt pronouns in Spanish.

Previous research has shown that across offline and online measurements, null pronouns often show robust effects and clearer subject-antecedent preferences than overt pronouns (Keating et al., 2011, 2016). In offline tasks, overt pronouns have not consistently shown strong preferences for object antecedents, as predicted by a clean division-of-labor account. In our current design, pronouns appeared before two possible antecedents, where participants had to decide where to resolve the cataphoric dependency. Posited by Carminati (2002), the Position of Antecedent Strategy predicts a clean division of labor between pronoun types. Accordingly, null pronouns are predicted to favor subject antecedents, whereas overt pronouns favor object antecedents. Conversely, an impatient-parser account suggests that parsers may link pronouns to the first available antecedent instead of waiting for later object antecedents. Thus, we manipulated the gender and plurality of the pronoun, subject, and object antecedents. This design was intended to test whether offline judgments show patterns consistent with a division-of-labor account.

The results of our naturalness judgments are partially consistent with the PAS. In offline judgments, null pronouns showed sensitivity to subject match. Despite supporting evidence for a subject bias, overt pronouns did not pattern as expected by a theory predicting a strong division of labor. Statistically, overt pronouns showed no difference between Match and Mismatch conditions. Thus, we could not conclude that participants reliably resolved overt pronouns in favor of object antecedents. Although the Match-Mismatch difference appeared clearer for null pronouns than overt pronouns, the Pronoun Type × Subject Match interaction was not significant. Thus, the present results do not provide strong evidence for a clean division of labor between null and overt cataphors. Instead, our findings suggest that null pronouns show subject-match sensitivity, while overt pronouns show no reliable preference for object antecedents. This pattern suggests that overt pronouns may be more variable than null pronouns in offline judgments.

While our current results may be compatible with the impatient-parser account, our study is consistent with Keating et al. (2011), who found a statistically significant subject bias but did not demonstrate a neat division of labor. Their 2011 findings showed a weakened subject bias with no clear preference for object antecedents. They suggest that these results are not surprising, as previous studies on monolinguals described a tendency to accept subject antecedents for overt pronouns. Our offline results show a similar pattern with participants rating Null Match items higher than Null Mismatch items.

Although a view based on a neat division of labor predicts that overt pronouns resolve in favor of object antecedents, we take our data to suggest that overt pronouns may show more variability than their null counterparts. Crucially, we do not take this variability to suggest a lack of preference for object-antecedents. Instead, we draw on prior work suggesting that offline and online data can diverge. Previous work on pronoun resolution reveals that offline and online measurements differ. It has been suggested that offline measurements, such as naturalness judgments, may reflect final interpretations rather than direct evidence of online processing. In Keating et al. (2016), PAS-like effects emerged in online self-paced reading tasks but became less clear in offline measures. Crosslinguistically, Kwon and Lee (2026) showed a reversal in offline and online findings. Notably, PAS-like results appeared in offline judgments but did not replicate in their online measures. This suggests that offline judgments may capture distinct processes from online processing.

For this reason, we suggest that the present offline results be treated as a baseline and as motivation for online measurements of real-time prediction. Thus, we use our evidence to motivate a future SPR study. This online study will use the same 2×2 factorial design, manipulating Pronoun Type and Subject Match, to test whether sensitivity to these factors emerges differently in real-time processing.

Crucially, we draw on the prediction that if participants use pronoun forms to guide a predictive mechanism, mismatch effects should appear when upcoming antecedents do not match the expectations generated by null pronouns. In contrast, we expect a weaker mismatch effect in constructions where the overt pronoun mismatches its expected antecedent. Thus, mismatch effects should be larger for null pronouns in comparison to overt pronouns. This contrasts with a prediction of the impatient-parser account, which suggests a parser seeks to resolve dependencies as quickly as possible. This account would predict a mismatch effect for both pronoun forms when the upcoming antecedent does not match a subject-based expectation.

This follow-up study will include comprehension questions to encourage active participation. In addition, we will reuse the filler items from the naturalness judgment task. The preregistration for the planned self-paced reading study is available on OSF.

Moreover, while our current design focuses on monolingual Spanish speakers, the original study that motivated this project focused on heritage and late-learning populations. Further work contributing to the PAS and cataphora may be interested in testing whether these populations share the same sensitivity to cataphoric constructions. This would extend the current study to research with bilingual populations and whether the aforementioned groups share similar strategies in pronoun resolution. We also believe that our current work poses cross-linguistic implications for how speakers of *pro*-drop languages resolve ambiguity. Work across Spanish, Korean, and Catalan has shown variability and distinctive outcomes, suggesting substantial cross-linguistic variation. Continued work on the PAS and other *pro*-drop languages may provide additional information to a division-of-labor account. Such work could help explain why offline and online measures sometimes produce asymmetric results and contribute to a stronger theory of when the division of labor between null and overt pronouns emerges.

In sum, the results of the current offline judgment study are partially consistent with a division-of-labor account in Spanish cataphora. The offline data aligned with the predicted pattern for null pronouns, which showed sensitivity to subject match. In contrast, overt pronouns did not show the predicted object-antecedent preference, suggesting more variability in offline judgments. Overall, the current study serves as a preliminary baseline for further online research, where self-paced reading measures can test whether these same patterns emerge in real-time processing.

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